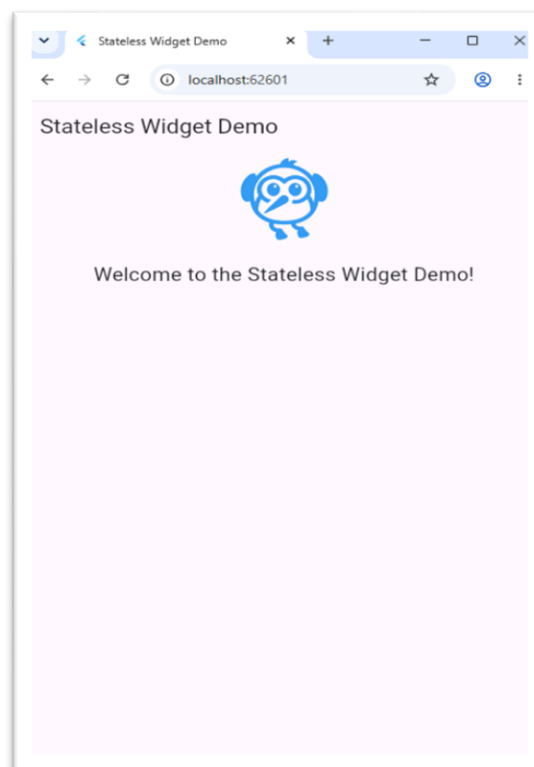


Practical -12: Design Flutter Application for Stateless and Stateful Widgets.

A **Stateless Widget** is a widget whose appearance does not change after it is built. It does not store or update any data during runtime. Stateless widgets are suitable for displaying fixed content such as text, images, icons, and labels.

Characteristics

- Immutable (cannot change after creation)
- Does not maintain state
- Displays static content
- Rebuilt only when its parent widget rebuilds
- Simple and efficient



main.dart

```
import 'package:flutter/material.dart';  
import 'pages/homepage.dart';
```

```
void main() {  
  runApp(const MainApp());  
}
```

```
class MainApp extends StatelessWidget {
```

```
const MainApp({super.key});

@override
Widget build(BuildContext context) {
  return const MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Stateless Widget Demo',
    home: HomePage()
  );
}
```

homepage.dart

```
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Stateless Widget Demo'),
      ),
      body: Center(
        child: WelcomeWidget(),
      ),
    );
  }
}
```

// Custom Stateless Widget

```
class WelcomeWidget extends StatelessWidget {
  const WelcomeWidget({super.key});
```

```
@override
Widget build(BuildContext context) {
  return Column(children: [
    Icon(Icons.flutter_dash, size: 100, color: Colors.blue),

    SizedBox(height: 20),

    Text(
      'Welcome to the Stateless Widget Demo!',
      style: TextStyle(fontSize: 20),
    )
  ],
);
}
```

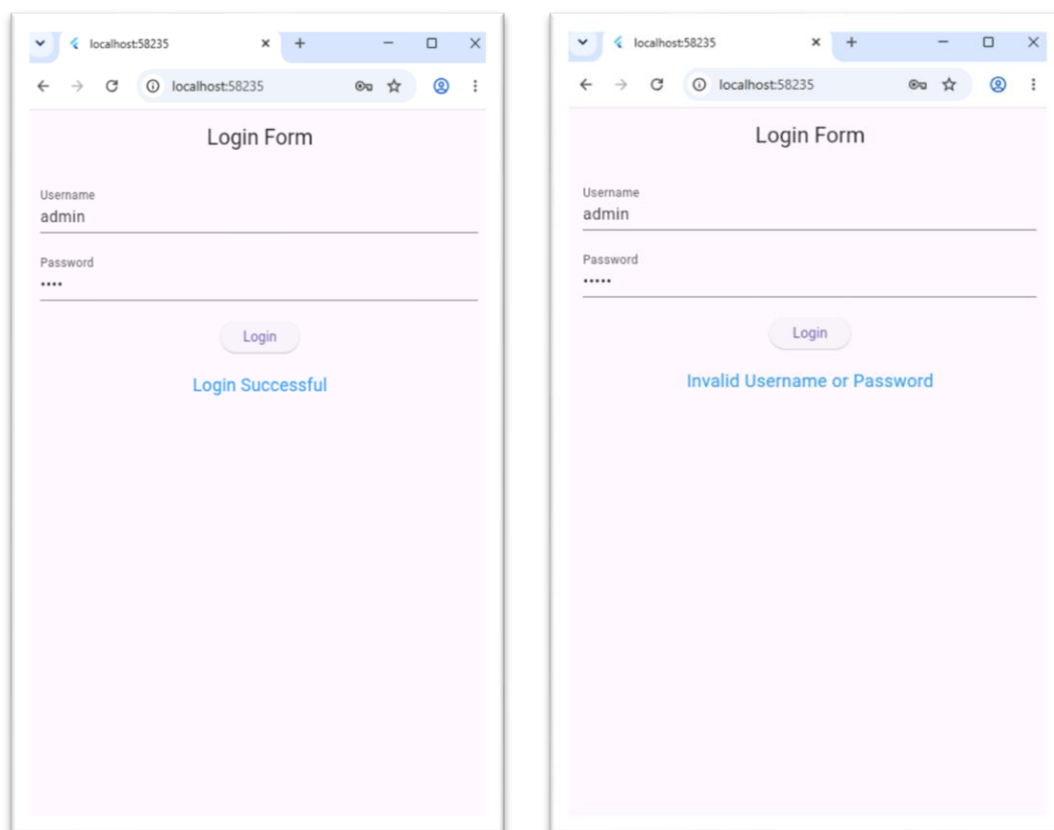
A **Stateful Widget** is a widget whose user interface can change during the execution of an application. It stores mutable data (state) and updates the UI whenever the state changes using the `setState()` method.

Characteristics

- Maintains mutable state
- UI changes dynamically based on user interaction
- Uses the `setState()` method to rebuild the widget
- Suitable for interactive applications

Examples:

- Counter App
- **Login Form**



main.dart

```
import 'package:flutter/material.dart';
```

```
void main() {  
  runApp(const MyApp());  
}
```

```
// Root Widget
```

```
class MyApp extends StatelessWidget {
```

```

const MyApp({super.key});

@override
Widget build(BuildContext context) {
  return const MaterialApp(
    debugShowCheckedModeBanner: false,
    home: LoginPage(),
  );
}

```

loginpage.dart

```

import 'package:flutter/material.dart';
// Stateful Widget
class LoginPage extends StatefulWidget {
  const LoginPage({super.key});

  @override
  State<LoginPage> createState() => _LoginPageState();
}

class _LoginPageState extends State<LoginPage> {

  String message = "";

  final TextEditingController usernameController =
    TextEditingController();

  final TextEditingController passwordController =
    TextEditingController();

  void login() {

    if (usernameController.text == "admin" &&
        passwordController.text == "1234") {

      setState(() {
        message = "Login Successful";
      });
    }
  }
}

```

```
} else {  
  
  setState() {  
    message = "Invalid Username or Password";  
  });  
  
}  
}
```

```
@override  
Widget build(BuildContext context) {  
  
  return Scaffold(  
  
    appBar: AppBar(  
      title: const Text("Login Form"),  
      centerTitle: true,  
    ),  
  
    body: Padding(  
      padding: const EdgeInsets.all(20),  
  
      child: Column(  
  
        children: [  
  
          TextField(  
            controller: usernameController,  
            decoration: const InputDecoration(  
              labelText: "Username",  
            ),  
          ),  
  
          const SizedBox(height: 20),  
  
          TextField(  
            controller: passwordController,  
            obscureText: true,  
            decoration: const InputDecoration(  

```

```
        labelText: "Password",
      ),
    ),

    const SizedBox(height: 20),

    ElevatedButton(
      onPressed: login,
      child: const Text("Login"),
    ),

    const SizedBox(height: 20),

    Text(
      message,
      style: const TextStyle(
        fontSize: 18,
        color: Colors.blue,
      ),
    )
  ],
),
),
);
}
```